Experiment No. 17 – Image Watermarking Using OpenCV

1. Aim

To design and implement a watermarking technique to effectively insert a watermark into the original image using OpenCV.

2. Program

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload original image
uploaded = files.upload()

# Read image
filename = list(uploaded.keys())[0]
image = cv2.imdecode(
    np.frombuffer(uploaded[filename], np.uint8),
    cv2.IMREAD_COLOR
)

if image is None:
    print("Error: Image not loaded")
else:
    # Create a copy of original image
    watermarked = image.copy()
```

```
# Watermark text
watermark = "COMPUTER VISION LAB"

# Get image dimensions
h, w = image.shape[:2]

# Set watermark position
font = cv2.FONT_HERSHEY_SIMPLEX
font_scale = max(0.5, w / 1000)
thickness = 2

text_size = cv2.getTextSize(
    watermark, font, font_scale, thickness
)[0]

x = w - text_size[0] - 20
y = h - 20

# Create overlay
overlay = watermarked.copy()

# Insert watermark text
cv2.putText(
    overlay, watermark, (x, y),
    font, font_scale, (0, 0, 255),
```

```
    thickness, cv2.LINE_AA
)

# Blend original and overlay
alpha = 0.4
watermarked = cv2.addWeighted(
    overlay, alpha, image, 1 - alpha, 0
)

# Display original and watermarked images
plt.figure(figsize=(12, 6))

plt.subplot(1, 2, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(cv2.cvtColor(watermarked, cv2.COLOR_BGR2RGB))
plt.title("Watermarked Image")
plt.axis("off")

plt.tight_layout()
plt.show()
```

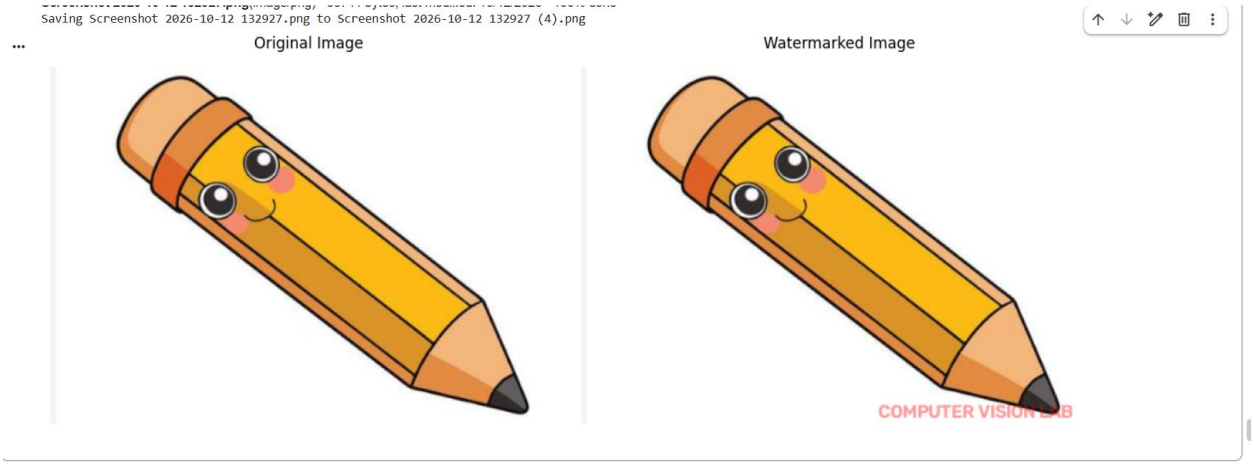
3. Expected Output

The output displays:

- Original Image.
- Watermarked Image with the text COMPUTER VISION LAB at the bottom-right corner.

The watermark appears partially transparent over the original image.

4. Output



The watermark text is successfully inserted into the original image using OpenCV and the add Weighted() blending function.

5. Result

Thus, the image watermarking technique was successfully implemented using Python and OpenCV.